

Lending Club Default Risk

EAS 587 Spring 2026 - Phase 4

Team

- Aayush Nepal
- Junwei Zhang
- Lusi Zhang

About this app

This single Streamlit app is both the team's live demo and the visual deliverable for Phase 4. Walk through the tabs left-to-right to see the full pipeline; the **Predict a loan** tab is the live model.

Project repository

Phases 1+2 received perfect scores;
Phase 3 not yet graded.

Welcome

Phase 1: Data

Phase 2: Pipeline

Phase 2: Models

Phase 3: Spark + Macro

Phase 3: MCP

Predict a loan

Insights + Next

Q&A

EAS 587 · SPRING 2026 · FINAL PRESENTATION

Lending Club Loan Default Prediction

Same use case, four implementations: pandas → scikit-learn → Spark + MCP → conversational UI

Aayush Nepal · Junwei Zhang · Lusi Zhang

The lender's question

Given a loan application, what is the probability this borrower will charge off rather than fully repay the loan?

We answer it end-to-end: filter and clean the public Lending Club dataset (2014-2017), train a tuned XGBoost model with strict leakage discipline, scale the data layer to Spark + Delta Lake, and deploy two ways - **as a conversational MCP tool inside Claude Desktop** and **as this interactive app** (Predict a loan tab).

At a glance

314,212

loans in the held-out 2017 test set

21.0%

actual charge-off rate

0.726

test AUC-ROC (XGBoost, full 2017)

110

preprocessed features fed to the model

How to navigate this presentation

1. **Walk the tabs left to right** to follow the project's evolution.
2. **Predict a loan** runs the actual production model live - try the preset selector for quick scenarios, then toggle compare mode to score two loans side-by-side.
3. The **Insights + Next** tab is the team's findings + future work.
4. **Q&A** is the closing surface for questions.

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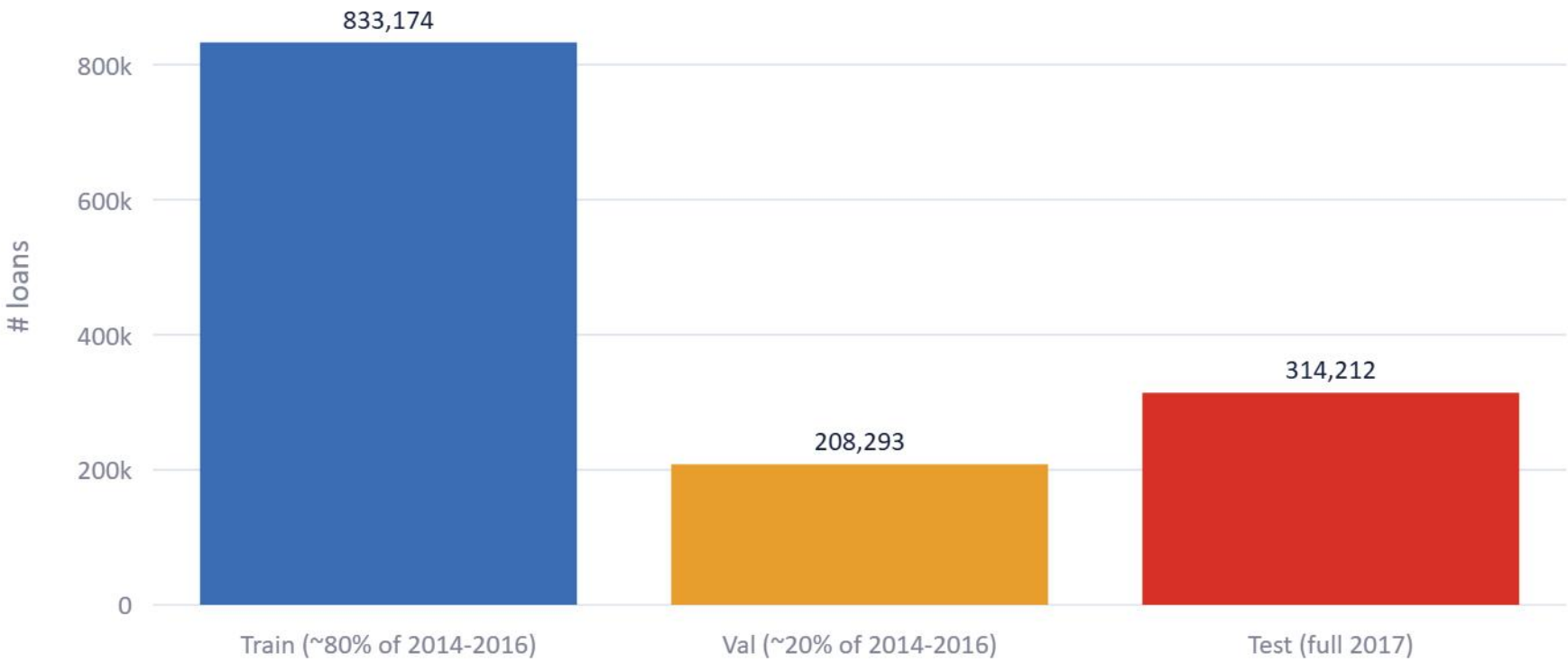
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PHASE 1

Data foundations

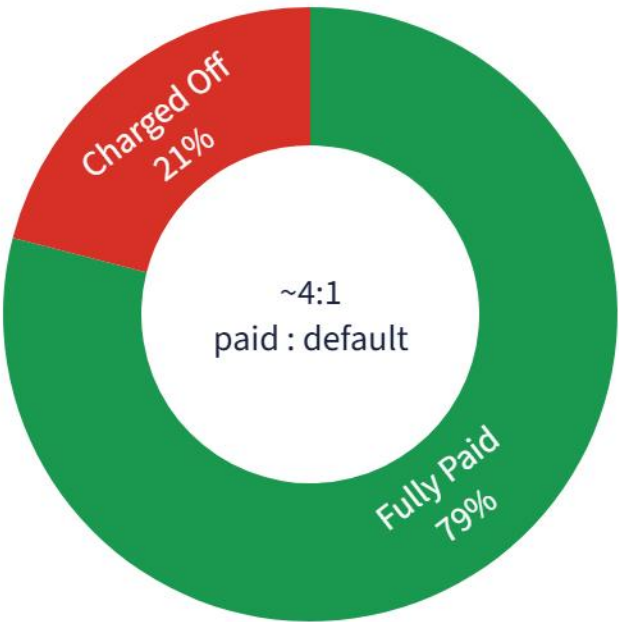
Filter, define the target, split with leakage discipline.

Time-based split (no random shuffling)



Train + val drawn from 2014-2016 by `issue_d` order; full 2017 held out as test. Random splits would leak the macro environment.

Class balance



We deliberately do NOT resample. Tree models handle the 4:1 imbalance natively via `scale_pos_weight`.

Leakage discipline - 21 post-loan columns dropped

21

post-loan columns dropped

These only get values **after** the outcome is known. Most public

Category	Columns dropped
Payment history	<code>total_pymnt*</code> , <code>total_rec_prncp/int/late_fee</code> , <code>last_pymnt_*</code>
Outstanding balance	<code>out_prncp</code> , <code>out_prncp_inv</code>

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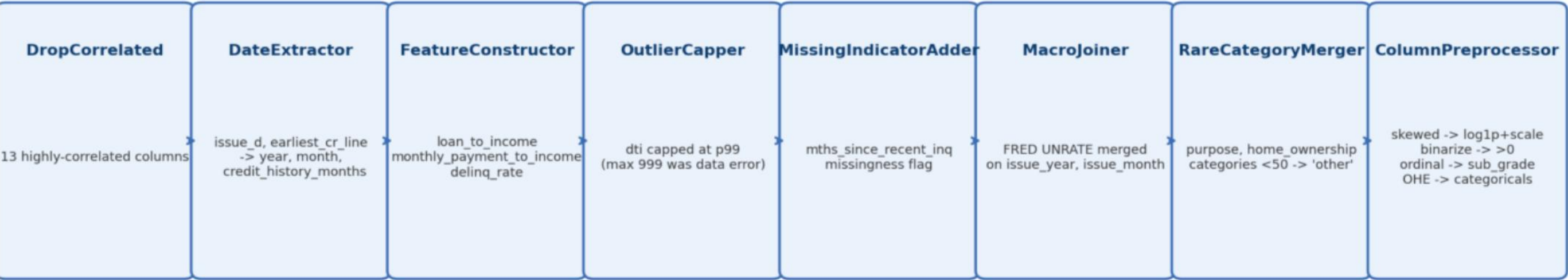
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PHASE 2

8-stage preprocessing pipeline

Custom scikit-learn transformers, fit on train only, applied identically to val and test.

Preprocessing pipeline - fit on train only, applied to val and test



Input: cleaned dataframe (49 columns)

Output: 110 features

Why each stage exists

DropCorrelated

13 highly-correlated columns removed (e.g. fico_range_high which is r=1.0 with fico_range_low).

DateExtractor

Splits issue_d and earliest_cr_line into year, month, and credit_history_months.

MissingIndicatorAdder

Some missingness is informative; we add a binary _missing flag for mths_since_recent_inq before imputing.

MacroJoiner

Pulls FRED UNRATE at fit time and joins on (issue_year, issue_month). Falls back to

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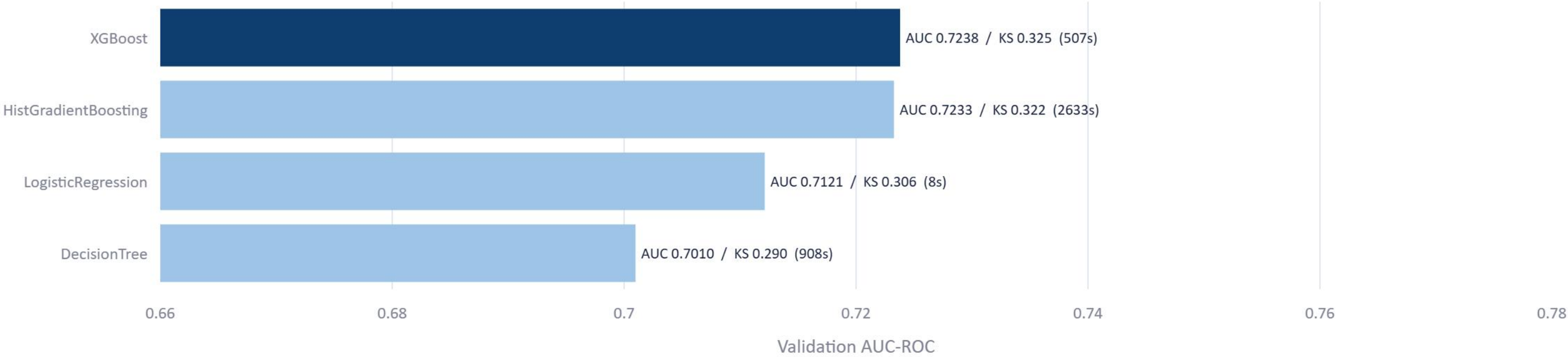
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PHASE 2

Model bake-off + holdout test results

Four algorithms, 50 Optuna trials each, MLflow tracked. XGBoost won; tested on the full untouched 2017 set.

Validation set bake-off



All four share the same preprocessing pipeline above. Tuning: Optuna TPE, 50 trials per model, optimizing val AUC-ROC. All 200+ trials logged to MLflow at `models/mlruns/`. Logistic Regression is the baseline (no tuning).

Held-out test results - XGBoost on all 314,212 2017 loans

0.7260

Test AUC-ROC

0.4103

Test AUC-PR

0.3279

Test KS

67.3%

Recall on defaults (at Youden threshold 0.495)

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PHASE 3

Spark, Delta Lake, and FRED macro overlay

Same data, same target, rebuilt on Databricks using the medallion architecture and extended with a FRED macro layer.

Medallion architecture - Bronze / Silver / Gold

BRONZE

Raw Kaggle archive landed as-is. Filtered 2014-2017 on Spark. Written to Delta as `bronze\_loans`.

SILVER

Cleaned and type-fixed. Leakage columns dropped. Schema enforced. -> `silver\_loans`.

GOLD

Time-based split + stratified train/val sample. Full 2017 test retained. -> `gold\_loans\_train/val/test`.

Secondary data: FRED macro overlay

The Phase 4 rubric specifically rewards adding recommended secondary data sources. We pulled 8 series directly from FRED (no auth) and engineered 4 more, then asked: does macro context improve the model?

National series

- UNRATE - civilian unemployment rate
- FEDFUNDS - effective federal funds rate
- CPIAUCSL - Consumer Price Index (all urban)
- CPI_YOY - 12-month CPI change (inflation proxy)

Engineered

- real_int_rate = int_rate - CPI_YOY
- rate_spread = int_rate - FEDFUNDS

State unemployment (top-5 LC volume states)

- CAUR , TXUR , NYUR , FLUR , ILUR
- state_unrate - per-borrower lookup based on addr_state

Production pipeline already uses macro

Phase 2's MacroJoiner injects unemployment_rate into every training row; the Phase 3 work tested whether richer macro (7 features) adds further lift.

Three macro insights

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PHASE 3

Conversational deployment via MCP

Same trained model, exposed as a Model Context Protocol tool. Claude Desktop can predict default risk from a plain-English loan description.

How it works

```
Claude Desktop
  | MCP (stdio transport)
  v
src/mcp/server.py  (FastMCP)
- loads preprocessing_pipeline.pkl + best_model.pkl on startup
- predict_loan_default():
  1. validates inputs
  2. builds a DataFrame (user inputs + 24 secondary defaults)
  3. PIPELINE.transform() - same steps as training
  4. MODEL.predict_proba() - returns risk + tier + recommendation
```

The pickle of the preprocessing pipeline references custom transformer classes under module path `__main__` (it was pickled while the script ran as `__main__`). Both this Streamlit app and the MCP server inject those classes into `__main__` before unpickling - same trick, two consumers.

Try it from Claude Desktop

Sample prompt

"Predict default risk for a \$12,000, 36-month loan at 11.5% interest. Sub-grade B2, \$58,000 income, DTI 15%, FICO 720, renting, debt consolidation."

What the tool returns

Default probability, risk tier (1-5), recommendation (Approve / Review / Decline), and the model name. Same scoring path as the ****Predict a loan**** tab here.

Setup details: see `src/mcp/README.md` and the `claude_desktop_config.json` template at `mcp_config_template.json` in the repo.

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Loan A

Quick preset ?

Custom

▼

Loan

Loan amount (\$)

15000.00

-

+

Term (months)

36

▼

Interest rate (%)

12.50

-

+

Sub-grade

B3

▼

Purpose

debt_consolida...

▼

Borrower

Annual income (\$)

65000.00

-

+

DTI (%)

18.00

-

+

FICO score

700

Home ownership

RENT

▼

Income verification

Not Verified

▼

Credit history

Open credit lines

10

-

+

Revolving utilization (%)

30.00

-

+

Total credit lines (ever)

20

-

+

Delinquencies (last 2 yrs)

0

-

+

Public derogatory records

0

-

+

Inquiries (last 6 mo)

0

-

+

Earliest credit line (YYYY-MM)

2010-01

Loan B

Quick preset ?

Custom

▼

Loan

Loan amount (\$)

15000.00

-

+

Term (months)

36

▼

Interest rate (%)

12.50

-

+

Sub-grade

B3

▼

Purpose

debt_consolida...

▼

Borrower

Annual income (\$)

65000.00

-

+

DTI (%)

18.00

-

+

FICO score

700

Home ownership

RENT

▼

Income verification

Not Verified

▼

Credit history

Open credit lines

10

-

+

Revolving utilization (%)

30.00

-

+

Total credit lines (ever)

20

-

+

Delinquencies (last 2 yrs)

0

-

+

Public derogatory records

0

-

+

Inquiries (last 6 mo)

0

-

+

Earliest credit line (YYYY-MM)

2010-01

Predict both loans

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streamlit.errors.StreamlitDuplicateElementId: There are multiple `plotly_chart` elements with the same auto-generated ID. When this element is created, it is assigned an internal ID based on the element type and provided parameters. Multiple elements with the same type and parameters will cause this error.

To fix this error, please pass a unique key argument to the `plotly_chart` element.

Traceback:

```
File "C:\Users\nepal\OneDrive\Desktop\College\dic\project\Eas587_project\src\app\streamlit_app.py", line 1299, in <module>
    render_prediction_panel(inputs_b, compact=True)

File "C:\Users\nepal\OneDrive\Desktop\College\dic\project\Eas587_project\src\app\streamlit_app.py", line 1204, in render_prediction
    st.plotly_chart(gauge_chart(proba, color), use_container_width=True,

File "C:\Users\nepal\OneDrive\Desktop\College\dic\project\Eas587_project\.venv\Lib\site-packages\streamlit\runtime\metrics_util.py",
    result = non_optional_func(*args, **kwargs)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\nepal\OneDrive\Desktop\College\dic\project\Eas587_project\.venv\Lib\site-packages\streamlit\elements\plotly_chart.p
    plotly_chart_proto.id = compute_and_register_element_id(
                             ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

File "C:\Users\nepal\OneDrive\Desktop\College\dic\project\Eas587_project\.venv\Lib\site-packages\streamlit\elements\lib\utils.py",
    _register_element_id(ctx, element_type, element_id)

File "C:\Users\nepal\OneDrive\Desktop\College\dic\project\Eas587_project\.venv\Lib\site-packages\streamlit\elements\lib\utils.py",
    raise StreamlitDuplicateElementId(element_type)
```

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